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Professional Summary

Robotics engineer focused on learning-based manipulation, humanoid locomanipulation, sim-to-real transfer, and real-world deployment. Experienced in building end-to-end robotic learning pipelines spanning simulation, policy training / distillation, vision integration, deployment software, and hands-on robot experimentation. Interested in robotic systems that actually create value.

Education

- **EPFL** — Lausanne, Switzerland
  M.Sc. Mechanical Engineering & Data Science; GPA: 5.5/6 — *2024.9 – 2027.5 (expected)*
- **ETH Zurich** — Zurich, Switzerland
  Exchange Student (Robotics); GPA: 5.6/6 — *2022.9 – 2023.2*
- **Seoul National University (SNU)** — Seoul, South Korea
  B.Sc. Mechanical Engineering; GPA: 3.9/4.3; summa cum laude — *2017.3 – 2024.2*

Publications

- Sim-to-Real Transfer for Manipulation with Zero-Shot Embodiment Augmentation via Mechanical Compliance
  - [**Jiwon You**, Cheng Pan, Sudong Lee, Josie Hughes] [Frontiers in Robotics and AI]

Selected Experience

- **ETH Zurich – Prof. Marco Hutter** — Zurich, Switzerland
  Master Thesis — *2026.9 – 2027.5*
  - **Perceptive humanoid loco-manipulation** on top of pretrained motion trackers, developing a modular architecture composed of high-level perceptive planner and low-level tracker.
  - Latent-space residual RL of kinematic tracker(SONIC) to enable physical interaction
- **EPFL – Prof. Josie Hughes** — Lausanne, Switzerland
  Research Assistant — *2025.7 – 2026.3*
  - Built a **sim-to-real manipulation pipeline** from scratch for a UR3 + UMI gripper using IsaacLab, covering simulation scene setup, policy training, and real-robot deployment.
  - Distilled state-based RL policies into an **end-to-end visuomotor policy** using behavior cloning, visual randomization, and a vision foundation model backbone
  - Developed a **Python deployment stack** for the full system, covering multithreaded robot control, Dynamixel communication and camera streaming.
  - Achieved **zero-shot transfer** on real hardware for object pickup from random targets in a 20 cm × 20 cm workspace without explicit state estimation.
  - **First author**, Frontiers in Robotics and AI
- **Switzerland Innovation Park Biel/Bienne** — Biel/Bienne, Switzerland
  Intern, Robotics / Sim-to-Real — *2026.2 – 2026.8*
  - Developed learning-based **industrial manipulation skills** for automated EV battery disassembly, focusing on contact-rich tasks such as screw engagement.
  - Trained and evaluated **variable stiffness impedance control policy** to improve tracking accuracy under joint stiction
  - Successfully delivered vision-guided object lifting demo with Kinova gen7 manipulator
  - Integrated vision foundation models(FastFoundationStereo, SAHI) to achieve better screw pose estimation
  - code maintenance, MDP design, streamlined policy deployment pipeline, hardware troubleshooting
- **EPFL RoboCup Team** — Lausanne, Switzerland
  Manipulation Software Engineer — *2024.9 – 2025.8*

- Built Python/ROS2 interfaces on top of the UFactory xArm SDK and Dynamixel SDK to provide reliable arm and gripper control primitives.
- Developed the **manipulation stack** for a mobile service robot, integrating **vision-based object detection and state estimation** with grasp planning and execution for real-world object pickup.
- Designed practical grasping heuristics and task-specific handling strategies to improve pickup robustness in real environments.

- **Zurich Robotics Hackathon 2025** Zurich, Switzerland
  *Autonomous Tic-Tac-Toe Robot **(link)*** *2025.5*
  - Built a fully autonomous robot system that played tic-tac-toe against humans in real life using a LeRobot SO-100 arm.
  - Developed the **LLM-based system layer** for gameplay reasoning, voice interaction, narration, and language-conditioning for the VLA policy

- **ERH Physical AI Hackathon 2026** Zurich, Switzerland
  *Vision-guided manipulation with contact-aware control **(link)*** *2026.9*
  - Built a system to pick up visually identical objects and distinguish mass from joint torque estimation
  - Deployed on OpenArm, compatible with Franka and UR via end-effector Cartesian control

- **Swiss Battery Technology Center** Biel/Bienne, Switzerland
  Research Intern *2025.2 – 2025.7*
  - Adapted an LLM-assisted reward-design workflow to fit internal robot learning workflows in IsaacLab.
  - Integrated the framework into reward tuning, curriculum tuning, and PPO hyperparameter optimization for manipulation tasks.
  - Applied the method to a contact-rich unscrewing task and benchmarked it against Ray Tune.

- **EPFL – Prof. Maryam Kamgarpour** Lausanne, Switzerland
  Research Intern *2025.9 – 2026.2*
  - Implemented a preference-based fine-tuning pipeline in simulation, from expert rollout generation and synthetic preference labeling to fine-tuning with DPO-style objectives and evaluation.
  - Studied stability trade-offs between pure DPO and mixed BC + DPO losses in control tasks; observed improved returns with the combined objective.

Skills

- **Robot Learning** – IsaacLab, RL workflows, sim-to-real transfer, policy distillation, behavior cloning, domain/visual randomization
- **Perception / Vision Integration** – vision-based object detection and state estimation, visuomotor policy learning with vision foundation model backbone
- **Robotics Software** – Python, PyTorch, ROS2, OpenCV, Docker, Git, Linux, C++
- **Hardware / Integration** – UR3, xArm, Franka, Kinova, OpenArm, multithreaded deployment interfaces, real-robot experimentation
- **Languages** – English (fluent), Korean (native)